ECI/ATM 149 Homework Assignment #1: Due Friday, 18 January 2019 by 3 pm

All HW assignments for ECI/ATM 149 must be turned in to the course HW Box located on the 2nd floor of Ghausi Hall. Remember that, as in every class you take at UCD, all incidents of plagiarism or cheating will be treated seriously and referred to student judicial affairs. You are required to adhere to the principles laid out in the **UC Davis Academic Code of Conduct** (http://sja.ucdavis.edu/cac.html). You must *show your work* in quantitative questions for full credit. *If you're work is not organized and legible it may not be graded.*

50 points total

1. Briefly explain the difference between a primary pollutant and a secondary pollutant and provide at least one example of each (3 pts).

2. Regional Emissions: Calculate (i) the total number of grams of primary $PM_{2.5}$ emitted from passenger cars over the course of an entire year in kg-$PM_{2.5}$ yr^{-1}, and (ii) the number of grams of primary $PM_{2.5}$ emitted on average per square meter per year (g-$PM_{2.5}$ m^{-2} yr^{-1}) in Yolo County from passenger automobiles. (This is termed an emissions *flux*.) (10 pts)

Given:

- Gasoline properties
  - Density = 0.72 g cm^{-3}
  - Carbon content = 0.842 g-C $(\text{g-fuel})^{-1}$ = 0.842 grams of carbon per gram of fuel
- Assume a fleet-average fuel economy of 24.7 miles gal^{-1} and that all passenger autos operate on gasoline (as opposed to diesel)[1]
- $EF_{PM2.5}$ = 0.05 g-$PM_{2.5}$/kg-fuel.[2]
- Yolo county area is 1,024 sq miles.[3]
- There are 121,083 registered autos in Yolo County.[4]
- Assume the average number of miles traveled annually per vehicle is 14,435.[5]
- Assume that Yolo people stay in Yolo (i.e. that all miles driven are in Yolo county).[6]

3. Emissions: The density of "woody biomass" in the Sierra Nevada forests is variable, but a typical value is 100 Mg per hectare.[7] A typical emission factor of $PM_{2.5}$ from a forest fire is 13.1 g-$PM_{2.5}$ per kg fuel.[8] The Camp Fire burned 153,336 acres.[9] If it is assumed that 50% of the biomass was consumed by fire, calculate the total amount of $PM_{2.5}$ emitted to the atmosphere from the Camp Fire (in kg). Compare

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[1] This is for the 2016 model year. The average for all on-road vehicles is certainly lower.
[2] Ban Weiss et al., *Atmos. Environ.* (2008); value is for primary $PM_{2.5}$
[3] https://en.wikipedia.org/wiki/Yolo_County,_California
[4] CA DMV
[5] http://www.carinsurance.com/Articles/average-miles-driven-per-year-by-state.aspx
[6] This is reasonable if we assume that the number of people leaving Yolo by car is balanced by the number of people entering Yolo by car each day.
[7] A Mg = megagram = 10^6 grams. A hectare = 100 acres. Reference is here.
[8] Reference: Andreae and Merlet (2001) (link)
[9] From Cal Fire (link)

your answer for this one fire to the estimated total $PM_{2.5}$ emitted from light-duty vehicles (aka passenger cars) for CA of 6.7 x 10^6 kg per year.[10] (6 pts)

4. Emissions: Reports from the 2015 VW diesel passenger vehicle scandal indicate that the vehicles emitted up to 35 times more NO_x than the relevant US standard of 0.07 g mi^{-1} = 0.043 g km^{-1} in CA.[11] There have been 489,000 diesel passenger cars sold in the US since 2009. There are approximately 250 million cars on the road in the US. If you assume that all of these diesel passenger vehicles emitted exactly 35 times as much NO_x as the US standard and that all other vehicles emitted at the standard limit, what would be the actual *fleet average* (aka weighted average) emission factor in g mi^{-1}? You can assume that the diesel vehicles are a subset of the entire fleet. (6 pts)

5. Emissions: **(a)** Use the EPA website and database of the National Emissions Inventory to determine what *fraction* of total NO_x emissions in the entire US in 2014 came from fuel combustion and what fraction came from mobile sources.[12] **(b)** What are the same fractions for California? **(c)** For CA, what are the fractions if fires are excluded as a source? **(d)** What types of sources does the "fuel combustion" category encompass? (9 pts total)

6. NAAQS: **(a)** How many criteria pollutants are currently regulated by the US EPA and what are they? **(b)** Use the EPA website to determine and report the largest *source sector sub-category* for each of the criteria pollutants on a national level and for California. (Hint #1: As an example of a "source sector sub-category", for CO on the national level the answer is "on-road mobile," being 37.09% of the total. Hint #2: for O_3 you can instead consider sources of volatile organic compounds, or VOCs.) (8 pts)

7. Based on the readings (see course schedule) and/or the links provided on Canvas (see the modules), identify at least one negative health impacts of exposure to O_3, one negative effect of O_3 on ecosystems and one negative effect of exposure to $PM_{2.5}$. (3 pts)

8. Emissions Sources: Provide an explanation as to why the primary sources of SO_2 and CO are so different (stationary vs. mobile) even though both come mainly from combustion of fossil fuels. (4 pts)

9. According to a recent study (summarized in an article available on Canvas here), outdoor air pollution contributes to 3.5 million premature deaths each year worldwide. What are the three main general sources identified, and list them in their order of importance (most-to-least). (Here "main source" refers to the broadest categories, and not the specific subcategories. So not, for example, "local waste disposal" but the broader source of which this is a subset.) (3 pts)

10. If the average carbon content of gasoline is 0.842 g-C (g-fuel)$^{-1}$ (from Problem 1), what must be the average chemical formula of the hydrocarbons that make up gasoline if it is assumed that gasoline is

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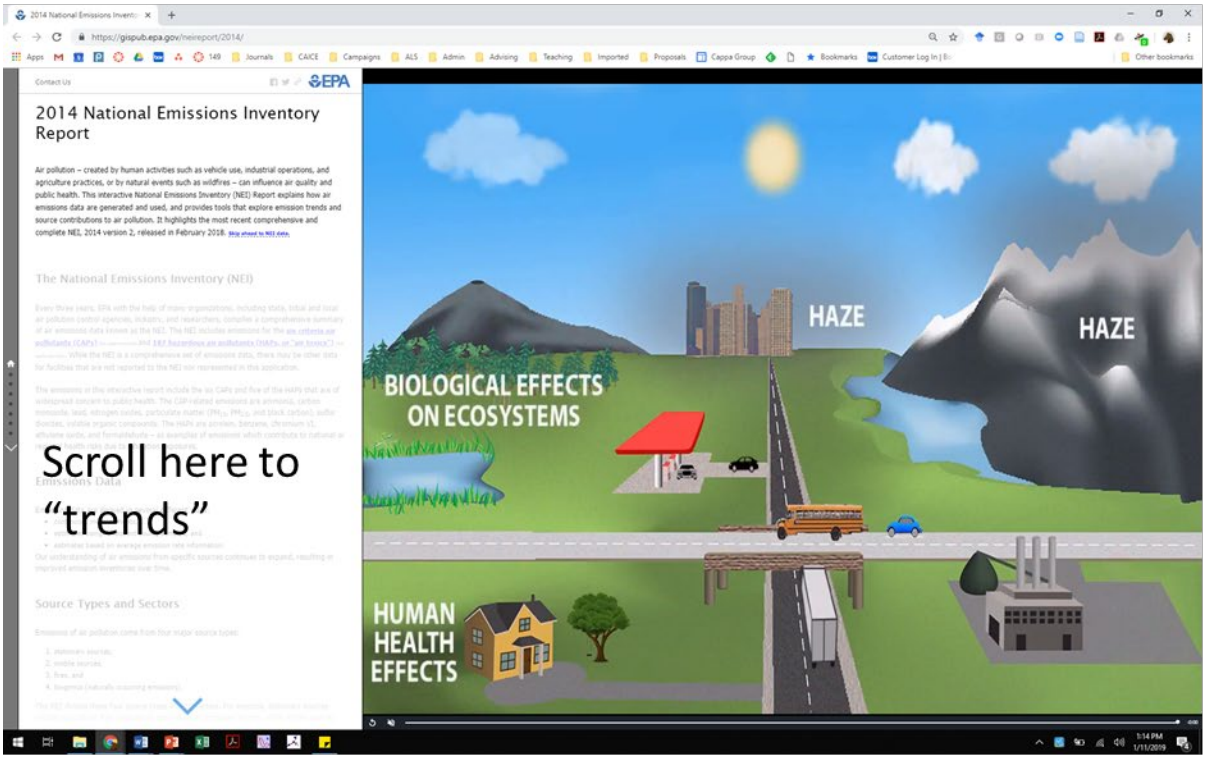
[10] Source: EPA

[11] See this link for the EPA standards. The diesel passenger vehicles are subject to Tier 2-Bin 5 exhaust standards.

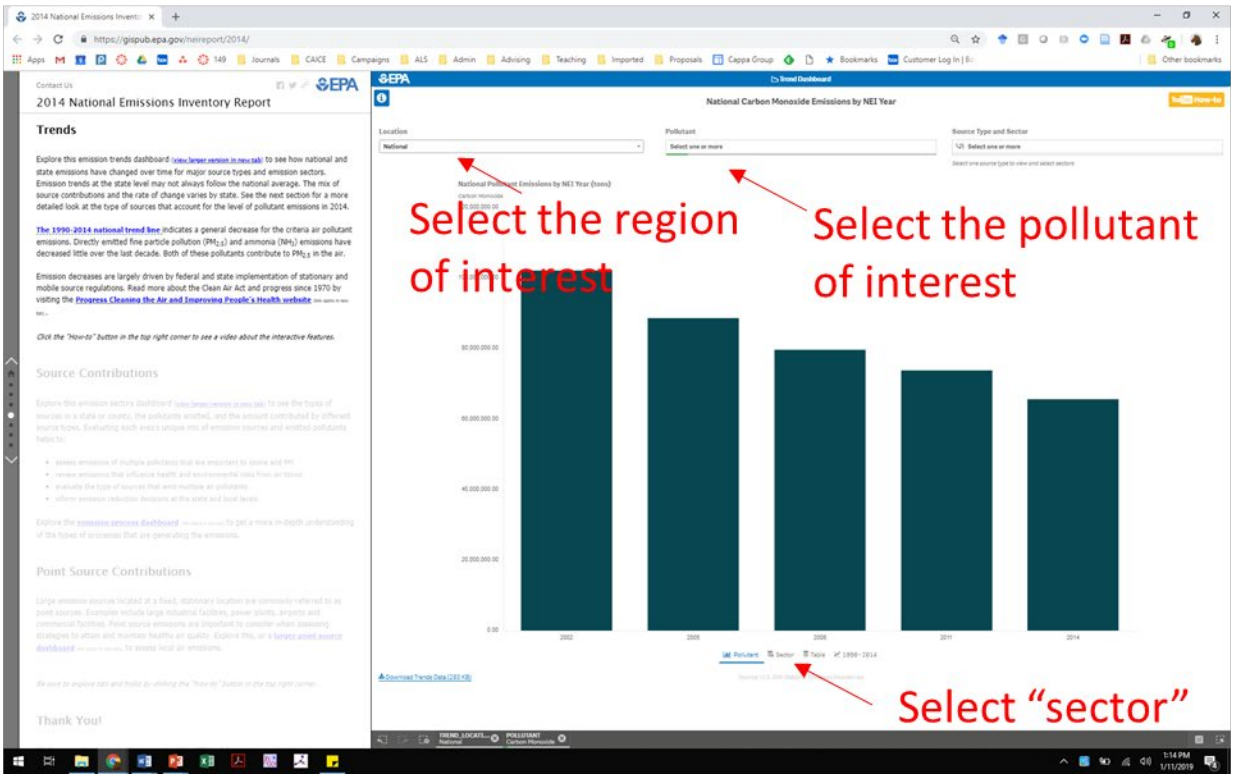
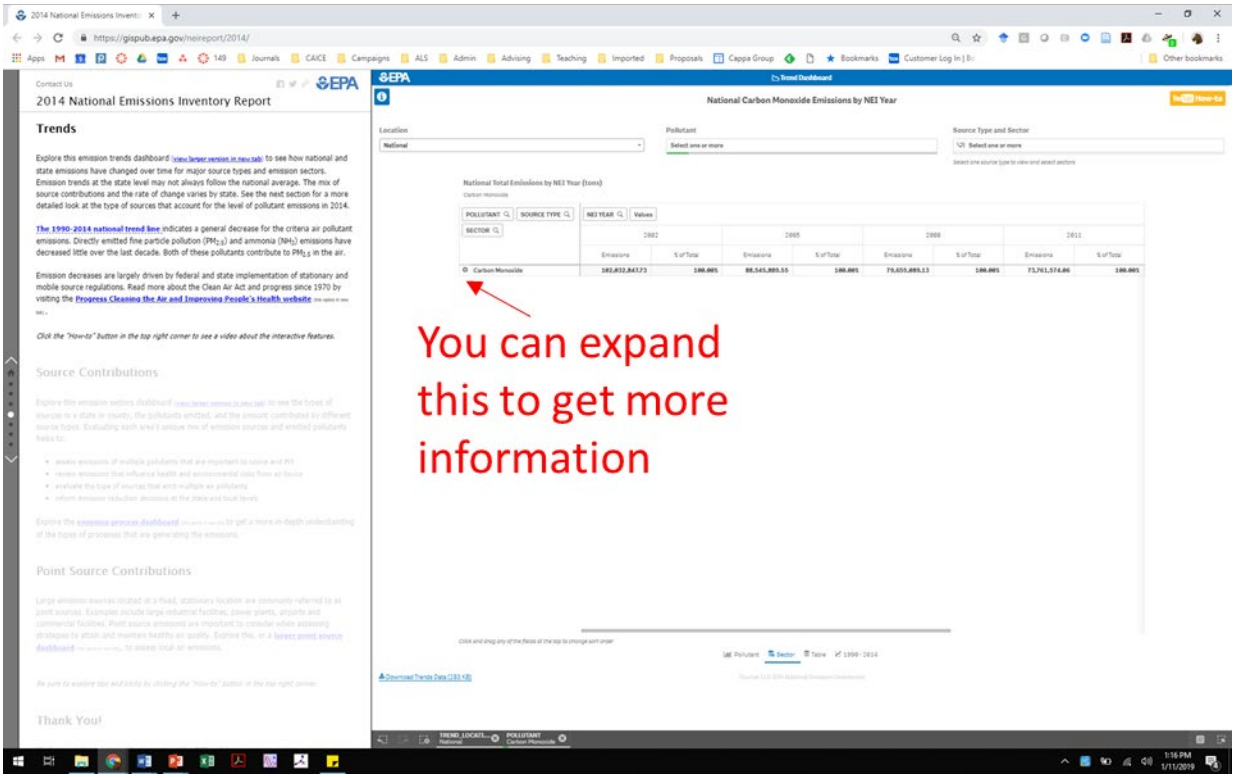
[12] Note that the EPA website is ***interactive***. You will need to scroll to the "Trends" section on the left, and then you'll interact with the graphs on the right. It helps to make the browser full screen. You'll want to click on the "Sector" designation below the graph that is likely shown by default. See the screen shots at the end of the document for guidance.

entirely composed of non-cyclic saturated hydrocarbons (i.e. molecules with generic formulas of C_xH_y)? (4 pts)

You will see this when you go to the page.



You should see something like this when you scroll to trends



When viewing by sector, you should see a table like this